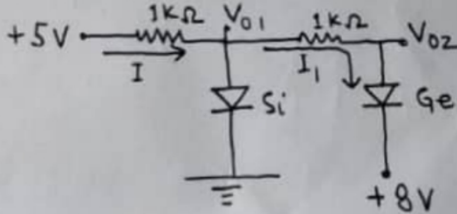


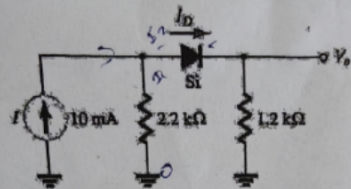
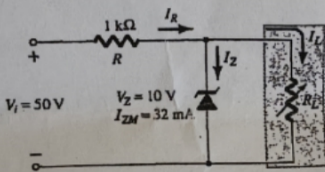
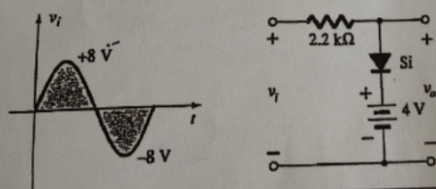
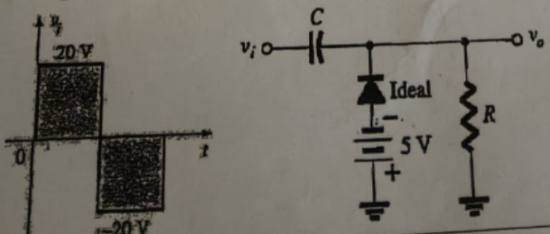
UNIT 1: P-N Junction Diode and its Applications

Syllabus topics: P-N Junction Diode, Depletion layer, Barrier potential, forward and reverse bias, Knee voltage, V-I Characteristics and its Equivalent Models, Avalanche and Zener Break Down, Diode Applications as Half Wave, Full Wave & Bridge Rectifier and their comparative analysis, Clippers, Clampers, Voltage Multiplier Circuit, Zener Diode and its Applications as a voltage regulator, Varactor diode.

Question 1

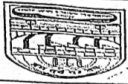
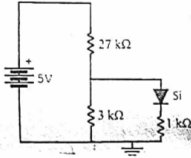
Harcourt Butler Technical University Kanpur		MID TERM-I		
Branch	CE, ME, EE, ET, CS, IT	Program	B.Tech	
Course Name	Introduction to Electronics Engineering	Semester	1	
Course Code	NET 101	Year	1 (2025-26)	
Time: 1:00 Hr	Answer All Questions	Maximum Marks	15	
Knowledge Level (KL)	K1: Remembering	K3: Applying	K5: Evaluating	
	K2: Understanding	K4: Analysing	K6: Creating	
Q.No	Questions	Marks	COs	KL
1.(a)	Explain the comparison of the conductivities of Insulator, Semiconductor and Metal on the basis of their energy band diagrams with appropriate diagram.	3	1	K2
(b)	The forward current through a Si diode is 10mA at room temperature. The corresponding forward voltage is 0.75 V. Find the value of reverse saturation current.	2	1	K3
2.(a)	Explain the reverse and forward bias operation of PN junction diode in brief, also draw the V-I characteristic curve of diode.	4	1	K1
(b)	Show that Ideal diode behaves like a switch, also draw its characteristic.	2	1	K4
3.	For given circuit find I , I_1 , V_{01} and V_{02} - 	4	1	K5
Course Outcomes	CO1	To understand the basic concept of diodes, and use the diode as a circuit element for different applications.		
	CO2	To understand the working of BJT, FET, OP-amp and their application.		
	CO3	To design the simple digital circuits using different logic gates.		
	CO4	To identify the errors while making electronic measurements and to understand the working of different types of transducers.		
	CO5	To understand the working principle of electronic instruments and displaying it on electronic devices.		

Question 2

Harcourt Butler Technical University Kanpur		First Mid Sem Exam		
Branch	BE/CH/PT/PL/OT/LT/FT/BioTech./BS-MS	Program	B.Tech	
Course Name	Introduction to Electronics Engineering	Semester	II	
CourseCode	NET 102	Year	2025-26	
Time: 1:00Hr	Answer All Questions	MaximumMarks	15	
Knowledge Level(KL)	K1:Remembering	K3:Applying	K5:Evaluating	
	K2:Understanding	K4:Analysing	K6:Creating	
Note: If any value is mission take appropriate value and solve the questions				
Q.No	Questions	Marks	COs	KL
1	Draw I-V characteristics of Ideal diode. Determine V_O and I_D in below circuit 	3	CO1	K1
2	For below Zener diode circuit, find the range of R_L and I_L . 	3	CO1	K4,5
3	Draw the output (v_o) waveform for given circuit- 	3	CO1	K3,5
4	Define Peak Inverse Voltage? For bridge rectifier value of the parameters (i) form factor, (ii) ripple factor, (iii) peak value & (iv) transformer utilization Factor.	3	CO1	K1,5
5	Draw the output (v_o) wave form for below figure- 	3	CO1	K4,5

FIRST MID SEM | NET 102 | BE/CH/PT/PL/OT/LT/FT/BioTech./BS-MS | 2025-26 | Q1 Ideal diode I-V characteristics — find V_O & I_D , Q2 Zener diode circuit — range of R_L & I_L , Q3 Output waveform for diode limiter circuit, Q4 Define PIV; bridge rectifier form factor/ripple factor/peak value/TUF, Q5 Output waveform for clamper circuit

Question 3

	Harcourt Butler Technical University Kanpur		FIRST SEM 202
Branch	CE/CSE/ET/IT/ME	Program	B.
Course Name	Introduction to Electronics Engineering	Semester	
Course Code	NET-101	Year	
Time: 1:00 Hr	Answer All Questions	Maximum Marks	
Knowledge Level (KL)	K1: Remembering	K3: Applying	K5: Evaluating
	K2: Understanding	K4: Analysing	K6: Creating
Note: Assume any data if necessary.			
Q. No	Questions	Marks	COs
1	Explain the working principle of a P-N junction diode under open circuit condition. Also discuss how the depletion region is formed.	3	CO1
2	Explain the working of a P-N junction diode under forward bias and reverse bias conditions along with the V-I characteristics.	3	CO1
3	Calculate the voltage (V_D) across the diode and the current (I_D) through the diode of Figure Q. 3. 	3	CO1
4	Differentiate between Avalanche breakdown and Zener breakdown in a P-N junction diode. Explain the mechanisms behind each type of breakdown and the conditions under which they occur.	3	CO1
5	Explain the operation of center-tapped full wave rectifier with the help of a circuit diagram. Define its PIV.	3	CO1

Course Outcomes	CO1	To understand the basic concept of diodes, and use the diode as a circuit element in different applications.
	CO2	To understand the working of BJT, FET, OP-amp and their application.
	CO3	To design the simple digital circuits using different logic gates.
	CO4	To identify the errors while making electronic measurements and to understand the working of different types of transducers.
	CO5	To understand the working principle of electronic instruments and display devices.

Ask Gemini



FIRST SEM MID SEM | NET-101 | CE/CSE/ET/IT/ME | Q1 PN junction diode under open circuit & depletion region formation, Q2 Forward & reverse bias PN diode with V-I characteristics, Q3 Diode circuit — find V_D and I_D , Q4 Avalanche vs Zener breakdown, Q5 Centre-tapped full wave rectifier operation & PIV

Question 4

	Harcourt Butler Technical University Kanpur			MID TERM-II
Branch	Electronics Engineering	Program	B.Tech	
And Course Name	Introduction to Electronics Engineering	Semester	I	
CourseCode	NET 101	Year	I (2025-26)	
Time: 1:00Hr	Answer All Questions	Maximum Marks	15	
Knowledge Level (KL)	K1: Remembering	K3: Applying	K5: Evaluating	
	K2: Understanding	K4: Analysing	K6: Creating	
Q.No	Questions	Marks	COs	KL
	Explain the mechanisms to explain the breakdown mechanisms in			

MID TERM-II | NET 101 | Electronics Engineering | 2025-26 | Q1 Mechanisms explaining the breakdown mechanisms in diode characteristic

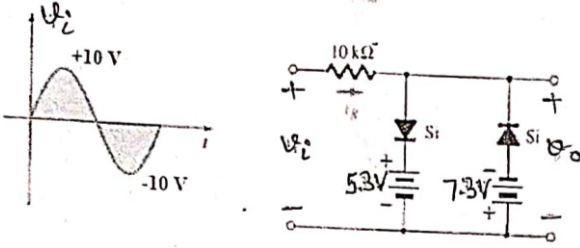
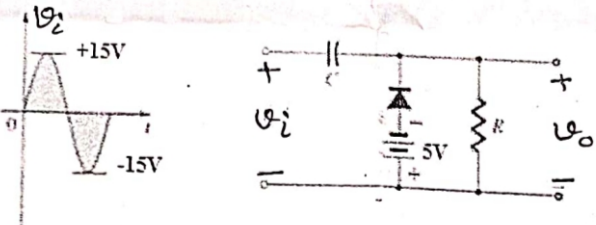
Question 5

For given circuit, $I_{Z(min)} = 5mA$, $P_{Z(max)} = 1Watt$, Find the range of R , such that load voltage is 8V constant.			
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Course Outcomes	CO1	To understand the basic concept of diodes, and use the diode as a circuit element for different applications.
	CO2	To understand the working of BJT, FET, OP-amp and their application.
	CO3	To design the simple digital circuits using different logic gates.
	CO4	To identify the errors while making electronic measurements and to understand the working of different types of transducers.
	CO5	To understand the working principle of electronic instruments.

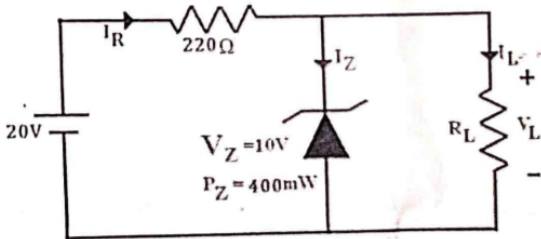
MID TERM-II | NET 101 | 2025-26 | Q4 Zener regulator circuit — find range of R such that load voltage remains constant at 8V

Question 6

		Harcourt Butler Technical University Kanpur		SEC SEM 202
Branch	CE/CSE/EE/ET/IT/ME		Program	B.
Course Name	Introduction to Electronics Engineering		Semester	
Course Code	NET-101		Year	
Time: 1:00 Hr	Answer All Questions		Maximum Marks	
Knowledge Level (KL)	K1: Remembering	K3: Applying	K5: Evaluating	
	K2: Understanding	K4: Analysing	K6: Creating	
Note: Assume any data if necessary.				
Q. No	Questions			Marks
1	Draw the output waveform for the sinusoidal input applied to the circuit shown in Figure Q. 1, by presenting all the necessary calculations which have been done to determine this output. 			3
Figure Q. 1				
2	Draw the output waveform for the sinusoidal input applied to the circuit shown in Figure Q. 2, by presenting all the necessary calculations which have been done to determine this output. 			3
Figure Q. 2				
3	Explain the zener diode as shunt (parallel) voltage regulator with fixed input voltage and variable load.			3
4	Explain working of BJT in CB configuration and sketch the input and output characteristics. Also indicate active region, cutoff region and saturation region on the output characteristic curve.			3

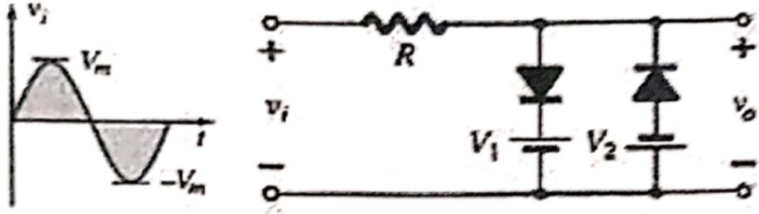
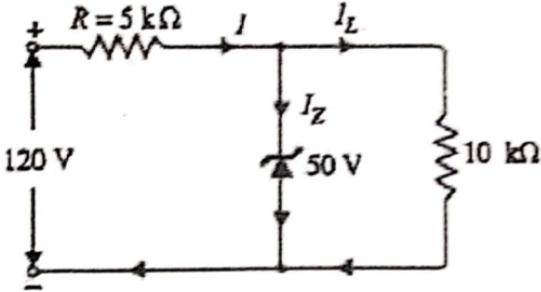
MID SEM-2 | NET-101 | CE/CSE/EE/ET/IT/ME | Q1 Output waveform for sinusoidal input through diode-clipper circuit (Fig. Q.1), Q2 Output waveform for sinusoidal input through diode circuit (Fig. Q.2), Q3 Zener diode as shunt (parallel) voltage regulator with fixed input and variable load

Question 7

	Harcourt Butler Technical University Kanpur		End Sem Exam 2024-25	
Branch	CE/CSE/EE/ET/IT/ME	Program	B. Tech	
Course Name	Introduction To Electronics Engineering	Semester	I	
Course Code	NET-101	Year	1 st Year	
Time: 2:30 Hr	Answer All Questions	Maximum Marks	50	
Knowledge Level (KL)	K1: Remembering	K3: Applying	K5: Evaluating	
	K2: Understanding	K4: Analyzing	K6: Creating	
Note: Assume any data if necessary. Attempt all parts of question at one place.				
Q.No	Questions	Marks	COs	KL
1(a)	A 230 V, 50 Hz A.C. is applied to the primary of a 5:1 step-down transformer in a bridge rectifier having a load resistor of 500 Ω assuming diodes are ideal diode. Find the following: (i) D. C. output voltage (ii) D. C. power at load (iii) PIV of each diode (iv) Output frequency	5	CO1	K1, K2
1(b)	Distinguish between Avalanche and Zener breakdown mechanism. Calculate I_R , I_Z , I_L and V_L for the network given below if $R_L=470\Omega$. 	5	CO1	K4, K5
2(a)	Explain the construction and working operation of N Channel Enhancement type MOSFET with suitable diagram. Also draw its drain	5	CO2	K1, K2

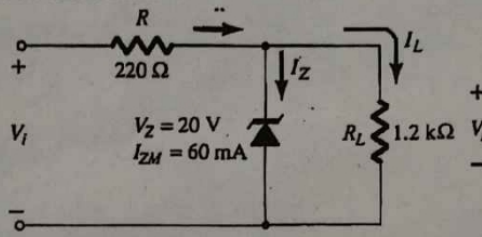
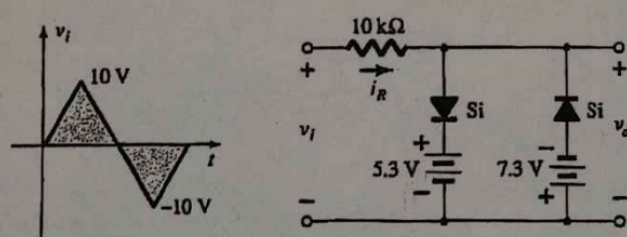
END SEM EXAM 2024-25 | NET-101 | CE/CSE/EE/ET/IT/ME | Q1(a) Bridge rectifier — DC output voltage, DC power, PIV, output frequency, Q1(b) Avalanche vs Zener breakdown; calculate I_R , I_Z , I_L , V_L for given network

Question 8

		Harcourt Butler Technical University Kanpur		Exa
Branch	CS/CE/EC/EE/IT/ME	Program	B.	
Course Name	Introduction To Electronics Engineering	Semester		
Course Code	NET101	Year	1	
Time: 2:30 Hr	Answer All Questions	Maximum Marks		
KnowledgeLevel (KL)	K1: Remembering	K3: Applying	K5: Evaluating	
	K2: Understanding	K4: Analysing	K6: Creating	
Q. No	Questions	Marks	CO	
1.(a)	Explain the working of center tapped full wave rectifier. Also calculate PIV .	3.5	1	
(b)	For the following circuit assuming both diodes are ideal -  Find the output waveform V_o , if $V_m = 9\text{V}$, $V_1 = 5\text{V}$ and $V_2 = 6\text{V}$	3.5	1	
(c)	For the given circuit find (i) Current through series resistance R (ii) Find zener current I_Z (iii) Find load current I_L . 	3	1	

END SEM PYQ's | NET101 | CS/CE/EC/EE/IT/ME | Q1(a) Centre-tapped full wave rectifier working + calculate PIV, Q1(b) Two ideal-diode circuit — find output waveform V_o , Q1(c) Zener circuit — find series current, zener current & load current

Question 9

Level(KL)	K2:Understanding	K4:Analysing	K6:Creating		
Note: If any value is missing take appropriate value and solve the questions					
Q.No	Questions	Marks	COs	KL	
1(a)	For below Zener diode circuit, find the range of V_i to maintain Zener diode in on condition.  Fig2.	5	CO1	K4,5	
1(b)	Draw the output (v_o) waveform for given circuit-  Fig 3	2	CO1	K3, 5	
1(c)	Explain the working of centre tapped rectifier. Also write the value of the parameters (i) form factor, (ii) ripple factor, (iii) peak value & (iv) transformer utilization Factor.	3	CO1	K1,5	
Q2(a)	Given that the $I_C = 2\text{mA}$ and $V_{CE} = 10\text{V}$, determine the value of R_1 and R_C for the below circuit given in Fig.4 . $\beta = 200$ 	5	CO2	K5	

Practice / Model Question Set (End Sem pattern) | NET-101 | Q1(a) Zener diode circuit — range of V_i to keep zener ON, Q1(b) Output waveform for triangular input through two-diode circuit, Q1(c) Centre-tapped rectifier — form factor, ripple factor, peak value, transformer utilization factor